

THE UNLOX ACADEMY

Weekly Assessment

SQL Foundations

Weekly Assessment

*DA / DS Track***Total marks: 100 Questions: 40****INSTRUCTIONS**

Complete all 40 questions.

The assessment uses a fresh dataset called flipcart.products. Run flipcart_setup.sql in MySQL Workbench BEFORE starting the assessment.

Section A (Theory, 8Q × 2.5) – write your answer letter (a/b/c/d) or short response.

Section B (Output Prediction, 8Q × 2.5) – describe what the query returns (row count, values).

Section C (Applied SQL, 24Q × 2.5) – write the actual SQL query. Formatting counts.

You may use Workbench to test your queries. You may NOT use ChatGPT / Copilot / any AI assistant. Any AI-detected answer receives zero for the entire section.

Submit as one .sql file or Word doc. File name: Week3_Assessment_<YourName>.

The Dataset - flipcart.products

A fictional Indian e-commerce catalogue. One table, 30 rows, 10 columns:

Column	Type	Notes
product_id	INT	Primary key, 1-30
product_name	VARCHAR(80)	e.g. 'Sony Bravia 55 TV', 'boAt Airdopes 141'
category	VARCHAR(30)	Electronics, Apparel, Home, Books, Beauty, Sports
brand	VARCHAR(30)	e.g. Sony, Nike, Prestige, Nykaa
price	DECIMAL(10,2)	Range: ₹249 to ₹79,999
stock_quantity	INT	Range 0 to 500. 2 products at 0 (out of stock).
is_active	BOOLEAN	1 discontinued product (is_active = FALSE)
launched_date	DATE	Range: 2017 to 2024
avg_rating	DECIMAL(3,2)	Range: 3.80 to 4.80. 3 products have NULL (new launches)

Column	Type	Notes
discount_pct	INT	0 to 40

Sample rows

product_id	product_name	category	brand	price	stock
active	rating	disc			
-----	-----	-----	-----	-----	-----
-- -----	----- -----				
1	Sony Bravia 55 TV	Electronics	Sony	65000.00	12
T	4.50 15				
2	Apple iPhone 15	Electronics	Apple	79999.00	25
T	4.80 5				
6	OnePlus Nord Buds 2	Electronics	OnePlus	2199.00	180
T	NULL 25				
8	Nokia 105	Electronics	Nokia	1499.00	500
F	3.80 0				
22	Rich Dad Poor Dad	Books	Kiyosaki	299.00	180
T	4.40 30				
27	WOW Skin Vitamin C Serum	Beauty	WOW	899.00	0
T	4.50 40				

(30 total rows. Data breakdown: Electronics 8, Apparel 6, Home 5, Books 4, Beauty 4, Sports 3.)

Section A - Theory

Choose the correct option (a, b, c, or d) for each question. Write only the letter.

A1. In MySQL, what is the relationship between a schema and a database?

- a) Schema is a group of databases
- b) Database is a group of schemas
- c) Schema and database are the same thing in MySQL
- d) Schema is a virtual view of a database

Your answer: _____

A2. Which SQL command family does TRUNCATE TABLE belong to?

- a) DML (Data Manipulation Language)
- b) DDL (Data Definition Language)
- c) DQL (Data Query Language)
- d) DCL (Data Control Language)

Your answer: _____

A3. You are designing a table to store product prices for an e-commerce site (e.g. ₹299.50, ₹79,999.00). Which data type is the most appropriate?

- a) FLOAT
- b) DECIMAL
- c) VARCHAR
- d) INT

Your answer: _____

A4. You try to INSERT a row into an orders table with a customer_id that doesn't exist in the customers table. There is a foreign key constraint. What happens?

- a) The row is inserted with NULL for customer_id
- b) The row is inserted and a warning is logged
- c) The INSERT fails with a foreign key constraint error
- d) The row is inserted but marked as invalid

Your answer: _____

A5. Which of the following statements is TRUE about WHERE vs HAVING?

- a) WHERE and HAVING can be used interchangeably
- b) WHERE filters individual rows before aggregation; HAVING filters groups after aggregation
- c) HAVING is used for row-level filtering; WHERE is used for group filtering

d) HAVING is faster than WHERE

Your answer: _____

A6. What is the result of the query: `SELECT * FROM products WHERE avg_rating = NULL;`

a) Returns all rows where avg_rating is NULL

b) Returns all rows where avg_rating is NOT NULL

c) Returns 0 rows (no rows match)

d) Throws a syntax error

Your answer: _____

A7. What is the difference between `COUNT(*)` and `COUNT(column_name)`?

a) They are identical

b) `COUNT(*)` counts all rows; `COUNT(column_name)` counts only non-NULL values in that column

c) `COUNT(*)` is faster but less accurate

d) `COUNT(column_name)` returns only distinct values

Your answer: _____

A8. Which command permanently removes an entire table (both structure and data), and CANNOT be undone by ROLLBACK?

a) `DELETE FROM table_name`

b) `TRUNCATE TABLE table_name`

c) `DROP TABLE table_name`

d) `REMOVE TABLE table_name`

Your answer: _____

Section B - Output Prediction

For each query below, describe what it returns. Give the row count and/or specific values.

B1. What does this query return?

```
SELECT COUNT(*) FROM products WHERE category = 'Electronics';
```

Your answer:

B2. How many rows does this query return?

```
SELECT * FROM products WHERE price BETWEEN 1000 AND 3000;
```

Your answer:

B3. What is the output of this query?

```
SELECT product_name FROM products  
WHERE category = 'Books' AND price < 400  
ORDER BY price DESC  
LIMIT 1;
```

Your answer:

B4. How many rows does this query return?

```
SELECT * FROM products WHERE avg_rating IS NULL;
```

Your answer:

B5. What single value does this query return?

```
SELECT MAX(price) FROM products WHERE category = 'Books';
```

Your answer:

B6. What does this query return?

```
SELECT category, COUNT(*) FROM products  
WHERE is_active = TRUE  
GROUP BY category  
HAVING COUNT(*) > 4;
```

Your answer:

B7. What does the output look like?

```
SELECT product_name,  
       CASE  
         WHEN price < 500 THEN 'Budget'  
         WHEN price < 5000 THEN 'Mid'  
         ELSE 'Premium'  
       END AS tier  
FROM products WHERE category = 'Beauty';
```

Your answer:

B8. What does this query return?

```
SELECT product_name, COALESCE(avg_rating, 0) AS rating  
FROM products  
WHERE stock_quantity = 0;
```

Your answer:

Section C - Applied SQL

Write the actual SQL query for each question. Formatting counts - use uppercase for keywords and align clauses on new lines for complex queries.

C1 - Basic SELECT + WHERE + ORDER BY

C1. Write a query to display all products with all columns.

Your query:

C2. Write a query that shows the product_name and price of all Books.

Your query:

C3. Write a query to list all products priced above ₹10,000, sorted from highest price to lowest.

Your query:

C4. Write a query to display the top 5 most expensive products in the Electronics category (show product name and price).

Your query:

C2 - Filtering Variations (IN, BETWEEN, LIKE)

C5. Write a query to list all products in either Electronics or Apparel category. Use the IN operator.

Your query:

C6. Write a query to find all products with a price between ₹500 and ₹2,000 (inclusive).

Your query:

C7. Write a query to find all products whose product_name contains the word 'Watch'.

Your query:

C8. Write a query to find all products whose brand starts with the letter 'S'.

Your query:

C3 - DISTINCT + Aggregate Functions

C9. Write a query to list all the unique categories in the products table.

Your query:

C10. Write a query to find the total number of products in the entire catalogue.

Your query:

C11. Write a query to find the average price of all Books.

Your query:

C12. Write a single query that returns the maximum and minimum price across all products.

Your query:

C4 - GROUP BY

C13. Write a query to count how many products are in each category.

Your query:

C14. Write a query to show the total stock quantity for each category.

Your query:

C15. Write a query to show the average price per category, sorted from highest average to lowest.

Your query:

C16. Write a query to show the count of products and the average price for each brand, showing only brands that have more than 1 product.

Your query:

C5 - HAVING + LIMIT

C17. Write a query to find categories that have more than 4 active products.

Your query:

C18. Write a query to find the 3 most expensive products in the entire catalogue.

Your query:

C19. Write a query to find all categories where the average price is above ₹2,000.

Your query:

C6 - NULL Handling

C20. Write a query to list all products where the avg_rating is missing.

Your query:

C21. Write a query to show product_name and rating for all products. If avg_rating is NULL, display the text 'New Launch' instead.

Your query:

C7 - CASE WHEN

C22. Write a query to display each product with a price_tier column classified as 'Budget' if price < ₹1,000, 'Mid' if price < ₹10,000, or 'Premium' if price >= ₹10,000.

Your query:

C23. For each category, write a query that shows the total product count and the count of 'Premium' products (price >= ₹10,000). Use SUM(CASE WHEN...).

Your query:

C24. Write a comprehensive query showing for each category (only categories with at least 3 products): total product count, count of active products, average price, and a category_tier column ('Cheap' if avg < ₹1500, 'Standard' if avg < ₹10000, 'Luxury' if avg >= ₹10000). Sort by average price descending.

Your query:

Submission

- **File name:** Week3_Assessment_<YourName>.sql
- **Upload to:** The Google Form. Deadline 7 days.
- **Integrity:** This is your work only. AI assistance, copy-paste from classmates, or answers from tutorial sites = zero and re-submission required.

All the best.
— The Unlox Academy